Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Process Digital Twin–Enabled Compression Molding of Composite Impellers with Integrated Blade Structures**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Smart Manufacturing*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Composite molding systems*	*Digital manufacturing*	*Process simulation / digital twin*
Concept of Proposed Technology	**Core Concept** The proposed technology aims to enable high-precision, scalable, and sustainable manufacturing of composite impellers through compression molding, supported by a process-level digital twin that predicts material flow, fiber orientation, weld-line formation, and the resulting structural performance of integrated impeller components [1,2]. Compression molding of complex impeller geometries inherently involves non-uniform material flow, spatially varying fiber orientation, and weld or knit line formation at flow convergence zones [1,2]. These process-induced features are strongly influenced by molding conditions such as pressure, temperature, charge placement, and molding sequence, and they play a critical role in determining the final mechanical and dimensional characteristics of the impeller.			

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	Key Functions and Potential Application To systematically address these process–structure interactions, the core concept of this technology is the development of a process digital twin that integrates material cards describing rheological behavior, fiber orientation evolution, and thermal properties with process models for compression molding flow, temperature evolution, and weld-line formation [3]. These models are coupled with process–structure relationships that link fiber architecture and weld-line location to impeller mechanical and dimensional performance [4]. By enabling simulation-guided tooling design and process parameter optimization, the proposed digital twin allows manufacturers to evaluate manufacturability and structural performance prior to fabrication, providing a predictive foundation for reliable production of high-performance composite impellers.
Rationale for Joint R&D Project	**Critical Technical Barriers** In current industrial practice, composite impellers are typically manufactured by separately producing blades and shroud components, followed by assembly using adhesive bonding and mechanical fastening. While this approach has enabled initial domestic production, it involves numerous manufacturing and assembly steps, high labor intensity, tolerance stack-up, and variability in quality, limiting scalability, dimensional accuracy, and long-term manufacturing efficiency. Although compression molding provides a promising pathway for simplifying composite impeller manufacturing through part consolidation, its successful application to complex rotating components cannot be achieved through empirical trial-and-error alone. Impeller geometries involve strongly three-dimensional flow around blades, rapid changes in local thickness, and multiple flow convergence zones, which introduce significant sensitivity to charge placement, flow paths, and pressure–temperature history. These effects become increasingly difficult to control consistently as manufacturing scales from laboratory trials to industrial production. At the root of this challenge is the heterogeneous nature of fiber-reinforced composite materials, in which the coupled flow of matrix and fibers governs the internal fiber architecture of the molded part. During mold filling and consolidation, fiber motion, reorientation, and interaction with complex tool geometries drive spatially varying fiber orientation, thickness variation, and the formation of weld or knit lines. These microstructural features directly determine the mechanical stiffness, strength, fatigue resistance, dimensional stability, and balance of the final impeller component, making predictive modeling essential for reliable manufacturing of high-performance rotating structures [1,2,4]. **Why Global Collaboration?** A joint R&D framework is therefore essential to bridge this gap between fundamental process understanding and industrial manufacturing practice. The Composites Manufacturing and Simulation Center (CMSC) at Purdue University is a globally recognized leader in digital twin–based simulation platforms for composite manufacturing, with extensive experience across multiple processes, including compression molding, resin-based processes, and additive manufacturing. By combining CMSC's expertise in process–structure modeling and digital twin development with Korean SMEs' capabilities in industrial tooling, compression molding operations, inspection, and impeller

	manufacturing, the proposed collaboration ensures that predictive models of flow, fiber orientation, and weld-line behavior are both scientifically robust and directly applicable to scalable industrial production.
Key Words	● Compression Molding of Fiber-Reinforced Composites ● Process Digital Twin ● Composite Flow and Fiber Orientation Modeling ● Weld / Knit Line Prediction ● Part-Consolidated Composite Structures

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

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The primary objective of this joint R&D project is to develop and validate a process digital twin–based framework for compression molding of composite impellers with integrated blade structures. Specific objectives include.

Detailed Objectives

- Developing material cards that capture flow behavior, fiber orientation evolution, and thermal properties of composite molding materials.
- Developing compression molding flow and thermal simulation models for complex impeller geometries.
- Predicting and controlling weld-line formation and fiber architecture in blade regions.
- Establishing process–structure–performance correlations for molded composite impellers.
- Demonstrating simulation-guided tooling design and process optimization to enable repeatable and reliable impeller manufacturing.

R&D Target Specifications

No	KPI	Unit	Target Value	Measurement and Verification Method
1	Degree of part consolidation (reduction in number of components)	%	≥ 40	Comparison of part count and assembly interfaces between baseline assembled impeller and integrated compression-molded impeller
2	Impeller dimensional accuracy (overall geometry)	mm	≤ ±0.5	3D scanning of molded impeller and comparison with CAD geometry
3	Fiber orientation prediction accuracy	%	≥ 85	Comparison between digital twin–predicted fiber orientation and experimental measurements (micrograph analysis or X-ray CT)
4	Weld / knit line location prediction accuracy	mm	≤ 5	Comparison between simulated weld-line locations and experimentally observed weld-line positions
5	Reduction in process development iterations	%	≥ 30	Comparison of number of physical trial runs required with and without digital twin–guided process optimization
6	Demonstration scale of molded impeller	-	Full-scale	Successful fabrication and validation of full-scale composite impeller
7	Validation in real application (electric train fan heater)	-	Pass	Functional performance, durability, and reliability testing in electric train fan heater system

R&D Plan

Year 1: Material Characterization and Digital Twin Foundation

The first year focuses on establishing the technical foundation for the process digital twin and baseline understanding of compression molding behavior for composite impellers.

- Material characterization of composite molding materials, including rheological behavior, thermal properties, and fiber orientation response under compression molding conditions.
- Baseline analysis of the current impeller manufacturing process to define reference geometry, assembly interfaces, and performance requirements.
- Development of initial compression molding flow and thermal simulation models for simplified

impeller geometries.

- Experimental studies on material flow, fiber orientation, and weld/knit line formation to support model calibration.
- Outcome: Validated material data and baseline digital twin capability enabling prediction of flow behavior and fiber orientation trends (supports KPIs 3 and 4; TRL 4).

Year 2: Digital Twin Enhancement and Simulation-Guided Process Design

The second year focuses on advancing the digital twin to handle full impeller geometries and using it to guide tooling and process design.

- Extension of flow, thermal, and fiber orientation simulations to full impeller blade geometries.
- Integration of weld/knit line prediction into the digital twin framework.
- Simulation-guided optimization of charge placement, pressure–temperature profiles, and molding sequence.
  Design of compression molding tooling for part-consolidated impeller blade–bottom structures based on digital twin results.
- Fabrication and testing of laboratory-scale and prototype impeller components to validate simulation predictions.
- Outcome: Demonstrated simulation–experiment correlation for fiber orientation and weld-line location, and reduced process development iterations through digital twin guidance (supports KPIs 1, 3, 4, and 5; TRL 5).

Year 3: Full-Scale Manufacturing, Validation, and Application Demonstration

The third year focuses on industrial-scale validation and real application demonstration.

- Fabrication of full-scale, part-consolidated composite impellers using optimized compression molding tooling and process parameters.
- Dimensional inspection, non-destructive testing (NDT), and mechanical performance evaluation of molded impellers.
- Final refinement of the digital twin based on full-scale manufacturing data.
- Integration of molded composite impellers into a representative electric train fan heater system.
- Functional performance, durability, and reliability testing under representative operating conditions.
- Outcome: Validated full-scale composite impeller manufacturing process and successful demonstration in a real railway application, confirming readiness for industrial deployment (supports KPIs 2, 6;).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Composites Manufacturing and Simulation Center (CMSC) at Purdue University will lead the development of the process digital twin framework for compression molding of composite impellers, with emphasis on material card development, digital tooling and process modeling, and process–structure–performance relationships governing manufacturability and structural reliability. CMSC will develop material cards capturing the mechanical, thermal, viscoelastic, and rheological behavior of composite molding materials, along with fiber orientation–dependent properties required for accurate structural performance prediction. These material cards will be integrated with digital tooling and process models that simulate compression molding flow in complex impeller geometries, including interactions with tooling geometry, pressure–temperature–time evolution, and prediction of fiber orientation and weld/knit line formation. Based on these coupled models, CMSC will establish process–structure–performance correlations linking molding parameters, tooling design, fiber architecture, and weld-line location to the mechanical and dimensional performance of molded impellers. CMSC will also lead simulation-guided tooling and process design, including evaluation of part-consolidated tooling architectures, charge placement strategies, and demolding feasibility. Laboratory-scale prototyping and controlled experimental validation will be conducted to calibrate and verify the digital twin, and standardized modeling workflows and design guidelines will be developed to ensure effective transferability to industrial composite impeller manufacturing.
Expected R&R of Korean SME Partner	The Korean SME partners will be responsible for industrial implementation, validation, and real-world demonstration of the developed compression molding and process digital twin technologies, ensuring practical applicability and commercialization readiness. Korean SME Partner will lead the industrial manufacturing and application of composite impellers while actively participating in part and tooling design in close collaboration with CMSC. Korean SME partner will co-develop impeller and tooling architectures informed by simulation results and industrial manufacturing constraints, ensuring manufacturability and application relevance. Korean SME partner will fabricate full-scale, part-consolidated composite impellers via compression molding using jointly developed tooling designs and simulation-guided process parameters, including pressure, temperature, and molding sequence. The company will conduct dimensional inspection and provide manufacturing feedback to support digital twin refinement, integrate molded impellers into functional assemblies, and supply manufacturing and performance data for validation and optimization of the digital twin framework, with a focus on railway applications. VASTEK will be responsible for inspection, durability evaluation, and validation of structural reliability for the manufactured composite impellers. This includes non-destructive testing (NDT) to identify internal defects, mechanical and failure testing under representative railway service conditions, and life-cycle and durability assessments. Where applicable, VASTEK will support validation testing in actual train operating environments and provide inspection, failure, and durability data to verify process digital twin predictions and long-term structural reliability.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** This joint R&D project will establish a scalable and reliable manufacturing framework for composite impellers enabled by process digital twin–guided compression molding and advanced tooling design. Composite impellers are critical rotating components used across strategic industries, including railway ventilation systems, shipbuilding and offshore applications, underwater vehicles, aerospace platforms, and unmanned aerial vehicles (UAVs), where they directly affect efficiency, reliability, noise, and operational safety. The developed manufacturing framework will replace assembly-intensive processes with a simplified and repeatable compression molding approach, improving dimensional accuracy, structural reliability, and manufacturing consistency while reducing development time, cost, and labor intensity. The integration of a process digital twin further enables simulation-guided design and process optimization, minimizing trial-and-error and supporting rapid customization for application-specific requirements. **Commercialization Potential** Currently, high-performance composite impellers are largely dependent on imported products, leading to long lead times, high costs, and limited design flexibility. This project directly supports domestic localization of advanced composite impeller manufacturing by enabling part-consolidated, high-precision production through compression molding, thereby reducing reliance on foreign suppliers and strengthening supply-chain resilience. **Industrial & Economic Impacts** Beyond railway applications, the proposed process and digital twin framework are readily transferable to impeller and rotor components for marine propulsion and ventilation systems, underwater platforms, aerospace thermal management, and drone propulsion systems. By establishing a validated domestic manufacturing capability, this project will enhance the global competitiveness of Korean SMEs, reinforce technological self-reliance, and create a foundation for expansion into high-value mobility, defense, and aerospace markets, supported by alignment with global engineering standards through collaboration with CMSC at research institute.

6. References

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